SMART CRACK DETECTION AND WATER INGRESS MONITORING IN RCC AND SFRC SPECIMENS USING SENSORS WITH ONLINE MONITORING

Niranjana K Vinod[1, a)], Rohith S[1, b)], Sana Santhosh[1, c)], Sneha R S[1, d)] and Divya Benny Mathew[1, e)]

[1] *Government Engineering College, Thrissur, Kerala, INDIA*

[a)] *Corresponding authors:* niranjanavinod124@gmail.com
rohith25ro@gmail.com
sanasanthosh314@gmail.com
snehars21@gmail.com
aswinch101@gmail.com

[b)] divyabmathew@gectcr.ac.in

Abstract: Crack formation and moisture ingress are critical factors influencing the durability and service life of reinforced concrete structures, while conventional inspection techniques remain inadequate for detecting early-stage damage and internal deterioration. This study presents an integrated structural health monitoring system for crack detection in Reinforced Cement Concrete (RCC) and Steel Fiber Reinforced Concrete (SFRC) beams using fiber optic sensors, with experimental evaluation carried out through three-point bending tests to analyse crack initiation and propagation behaviour. In addition, a controlled investigation of moisture ingress was conducted on concrete cube specimens using capacitance-based sensing techniques to understand moisture behaviour under varying conditions. The results indicate that the inclusion of steel fibers enhances crack resistance and improves crack distribution due to fiber-bridging mechanisms, while the moisture study establishes a relationship between permeability characteristics and dielectric response of concrete. The monitoring system is further integrated with an ESP32-based IoT platform to enable real-time data acquisition and visualization. The proposed approach provides a reliable framework for the combined assessment of structural performance and durability, contributing to the development of intelligent and sustainable infrastructure systems

Keywords: Structural Health Monitoring, Fiber Optic Sensors, Water Ingress, Steel Fiber Reinforced Concrete, IoT, ESP32

1. Introduction

Concrete is the primary material in civil engineering due to its versatility and high compressive strength. Reinforced concrete structures are inherently vulnerable to cracking due to the combined effects of mechanical loading, shrinkage, thermal stresses, and environmental exposure. The presence of cracks not only compromises structural integrity but also facilitates the ingress of water and aggressive agents such as chlorides and sulfates, thereby accelerating deterioration processes including reinforcement corrosion and loss of durability.

Conventional inspection methods, predominantly based on visual assessment, are limited in their ability to detect early-stage micro-cracks and internal moisture variations. As a result, critical damage often remains undetected until it reaches an advanced stage, leading to increased maintenance costs and potential safety risks.

In this context, Structural Health Monitoring (SHM) systems have emerged as a promising solution for continuous and real-time assessment of structural performance. Among the available sensing techniques, fiber optic sensors have gained significant attention for crack detection due to their high sensitivity to strain and micro-bending effects, which result in measurable variations in transmitted light intensity during crack initiation and propagation. Similarly, capacitance-based sensors employing conductive plates are effective for monitoring water ingress, as they detect changes in dielectric properties when water replaces air within the pore structure of concrete.

Simultaneously, the incorporation of advanced materials such as Steel Fiber Reinforced Concrete (SFRC) has demonstrated significant improvements in crack resistance, ductility, and post-cracking behaviour due to the fiber bridging mechanism. However, limited research has explored the combined application of advanced materials and real-time monitoring systems.

This study aims to evaluate an integrated monitoring system for RCC and SFRC beams to compare their structural performance and durability under simulated loading and moisture exposure, while also evaluating the performance enhancement offered by SFRC.

2. Experimental Program

The primary objective of this study was to investigate the effectiveness of an integrated sensor-based structural health monitoring system for detecting crack initiation and monitoring moisture ingress in concrete structures. For this purpose, an experimental programme was developed comprising RCC and SFRC beam specimens for crack

detection analysis and cube specimens for moisture ingress study. The beam specimens were instrumented with embedded fiber optic sensors positioned in the tension zone to capture strain variations associated with crack development. The beams were subjected to three-point bending tests under gradual loading conditions until failure.

In addition, a separate set of concrete cube specimens was prepared to analyse moisture ingress behaviour using capacitance sensors under controlled exposure conditions. The cube-based study was conducted to establish the relationship between moisture content and dielectric response, providing supporting insight into durability aspects. The monitoring system was further integrated with an ESP32-based IoT platform for real-time data acquisition and visualization.

2.1 Specimen Matrix and Geometry

Beam specimens were prepared for flexural crack detection, while cube specimens were used for moisture ingress studies. The beam specimens were reinforced with steel bars and transverse stirrups to ensure flexural failure, with careful detailing adopted to promote crack initiation in the tension zone and to facilitate effective sensor placement.The reinforcement cage consisted of two 10 mm diameter high-yield strength deformed (HYSD) bars provided in the tension zone, positioned near the bottom of the beam with a clear cover of 20 mm. Transverse reinforcement was provided in the form of two-legged 6 mm diameter stirrups spaced at 90 mm centre-to-centre along the beam length to resist shear and ensure controlled crack development.

The specimens were categorised into two experimental series to evaluate flexural crack behaviour and moisture ingress characteristics under controlled conditions:

1. Crack Detection**:** Six beam specimens, each measuring $1000\times100\times150\,mm$, were casted to study flexural behaviour. The series consisted of three conventional RCC beams and three SFRC beams. The SFRC specimens incorporated hooked-end steel fibers at a volume fraction (V_f) of 0.5% to enhance crack resistance and post-cracking behaviour through fiber-bridging action. Fiber optic sensors were embedded along the tension zone near the reinforcement to monitor crack initiation and propagation.

2. Water Ingress**:** Six cube specimens, each measuring $150\times150\times150\,mm$, were cast to investigate moisture ingress characteristics. This series comprised three SCC cubes and three SFRC cubes. A capacitance-based sensing approach was adopted to monitor moisture penetration by measuring changes in dielectric properties under controlled exposure conditions.

2.2 Materials and Mix Proportion

The study utilized a high-performance Self-Compacting Concrete mix design with a water-to-cement (w/c) ratio of 0.43. Ordinary Portland Cement (OPC) 53-grade was used as the primary binder. To achieve the necessary rheological properties for SCC - specifically high flowability and segregation resistance - a polycarboxylate ether-based superplasticizer, Auramix 450, was employed.

For the Steel Fiber Reinforced Concrete specimens, hooked-end steel fibers were incorporated at a volume fraction of 0.5%. This specific dosage was selected based on existing literature which indicates that a 0.5% fiber addition significantly enhances post-cracking ductility and micro-crack bridging without compromising the workability of SCC. The fresh properties of the concrete were validated through Slump Flow, V-Funnel and L-Box tests, while hardened properties were assessed through compressive strength tests.

Fig.1 Testing of Fresh Concrete Properties

Table 1 Properties of Concrete

Category	**Property**	**SCC**	**SFRC**
Fresh Concrete Properties	Slump Flow	780 mm	620 mm
	L-Box Ratio	0.80	0.65
	V-Funnel	12 sec	22 sec
Hardened Concrete Properties	Average Compressive Strength (7 Days)	27.6 N/mm^2	33.23 N/mm^2

	Average Compressive Strength (28 Days)	40.26 N/mm^2	51.6 N/mm^2

2.3 Sensor Installation and Instrumentation

The Smart functionality of the specimens was achieved through a dual-track sensor installation process targeting both mechanical strain and environmental ingress.

Fiber Optic Sensor (FOS) Integration: To monitor flexural cracking, fiber optic sensors were installed in the bottom tension zone of the beams. A set of three optical fibers was utilized for each specimen to ensure redundant and accurate strain measurement. Two distinct installation methods were employed:

- Direct Embedding: Fibers were positioned precisely 5 mm into the bottom concrete cover to capture surface-proximate micro-cracking.
- Reinforcement Coiling: One set of fibers was coiled along the longitudinal rebar to monitor the bond-slip behaviour and internal strain transfer during loading.
- An photo-diode based monitoring system was used to measure the intensity of the light though the fiber and detect changes in it

Fig.2 Sensor Arrangement

Capacitance-Based Moisture Monitoring: For the durability series, capacitance-based conductive plates were utilized to track water ingress. These plates were secured to the side surfaces of the cubes using a high-strength adhesive. The experiment involved a controlled water exposure protocol where 50 ml of water was applied at fixed time intervals. The sensors, interfaced with an ESP32-based astable multivibrator circuit, recorded the fluctuations in dielectric permittivity as moisture penetrated the concrete matrix.

Fig.3 Conductive plates for moisture detection

2.4 IoT System Architecture and Circuit Design

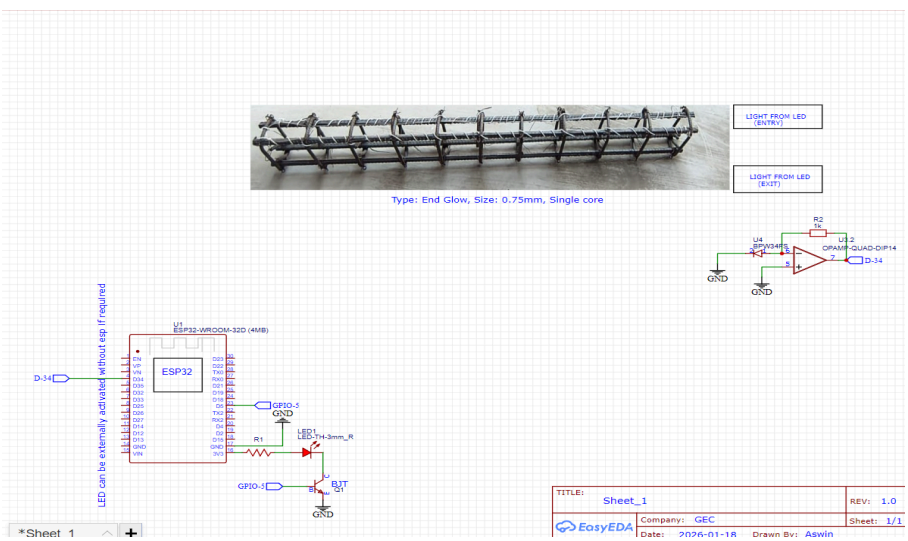

Fig.5 Crack detection circuit

change in intensity if light gets detected by a photo diode , a transimpedance amplifier converts current provided by the photodiode to voltage, the esp32 process the data and hosts the website

2.5 Testing Protocol

For crack detection: The beam specimens were subjected to a three-point bending test to correlate the FOS light intensity modulation with physical crack widths. Simultaneously, the cube specimens were monitored under controlled environmental conditions to quantify the rate of moisture ingress in SCC versus SFRC.

Fig.4 Beam testing with 3-point bending

For water ingress monitoring: The experimental monitoring of water ingress was performed through controlled incremental addition of water to 150 mm concrete cubes. Water was introduced in steps of 50 ml, and adequate time was allowed after each increment to ensure uniform capillary absorption before recording the sensor response. A parallel-plate capacitive sensor integrated with an ESP32 microcontroller was used to measure variations in capacitance arising from changes in the dielectric properties of concrete due to moisture ingress.

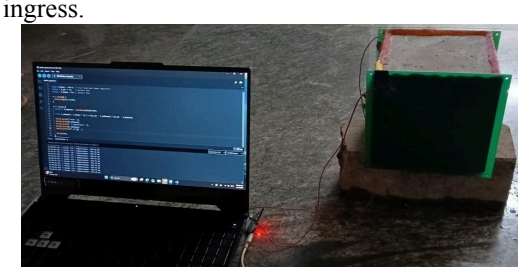

Fig.5 Water Ingress Monitoring Using Capacitive Sensor and ESP32 Interface

The touch sensor pins of esp32 was used to measure the change in capacitance of the concrete block, the plates were directly wired to the touch pin (gpio13) and ground, and data was displayed using the website hosted

3. Results and Discussion

The experimental investigation was carried out to evaluate the effectiveness of the developed sensing system in identifying crack initiation, propagation, and water ingress in concrete specimens. Reinforced concrete and steel fiber reinforced concrete beams were tested under flexural loading, and the sensor outputs (ADC values and dielectric response) were correlated with structural behaviour at different loading stages.

3.1 Crack Detection Behaviour

The variation of ADC values with applied load clearly indicated distinct stages of structural response, as shown in Fig. 6 (Load-ADC Curve for RCC Beam). In RCC beams, a stable and nearly linear ADC trend was observed within the safe load range of 0-20 kN, corresponding to elastic behaviour with no visible cracking.

At approximately 20 kN, a deviation from linearity was observed in the Load-ADC graph (Fig. 6), indicating crack initiation at the tension zone. With further increase in load to the range of 30-40 kN, a sharp reduction in ADC values was recorded, corresponding to rapid crack propagation and widening of a dominant flexural crack at mid-span. Near the ultimate load range of 48–52 kN, the graph shows an abrupt drop in ADC values, indicating sudden failure of the beam.

In contrast, the Load-ADC response of SFRC beams, presented in Fig.7, exhibited a more gradual and stable variation. The elastic range extended up to 25 kN, after which initial cracking was indicated by minor fluctuations in ADC values rather than a sudden drop.

During the crack propagation stage (33–43 kN), the graph (Fig. 7) shows multiple small variations, corresponding to distributed micro-cracking. This behaviour is attributed to fiber bridging, which controls crack widening and delays localization. Even near the ultimate load range of 48-52 kN, the ADC values decreased progressively, demonstrating enhanced ductility and improved post-cracking behaviour.

The comparison of Fig.6 and Fig.7 clearly demonstrates that RCC beams exhibit abrupt sensor response due to localized cracking, whereas SFRC beams show a distributed cracking pattern with gradual sensor variation. This confirms the sensitivity of the sensing system in distinguishing different structural behaviours.

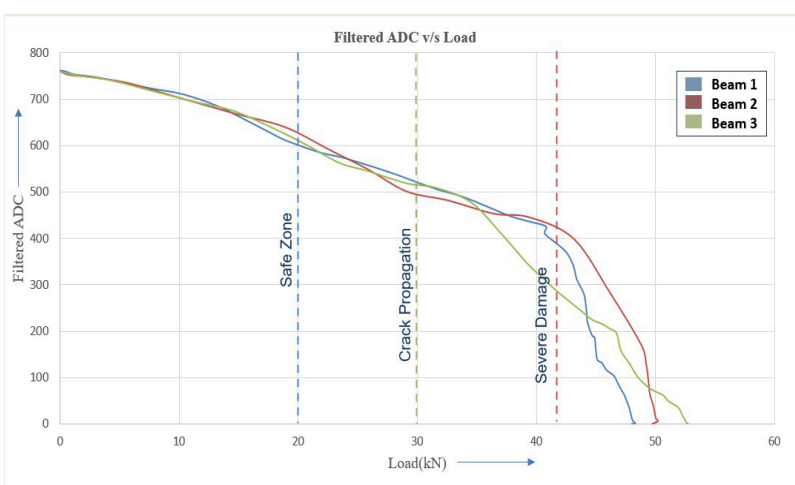

Fig.6 Graphical representation showing ADC vs Load relationship in RCC

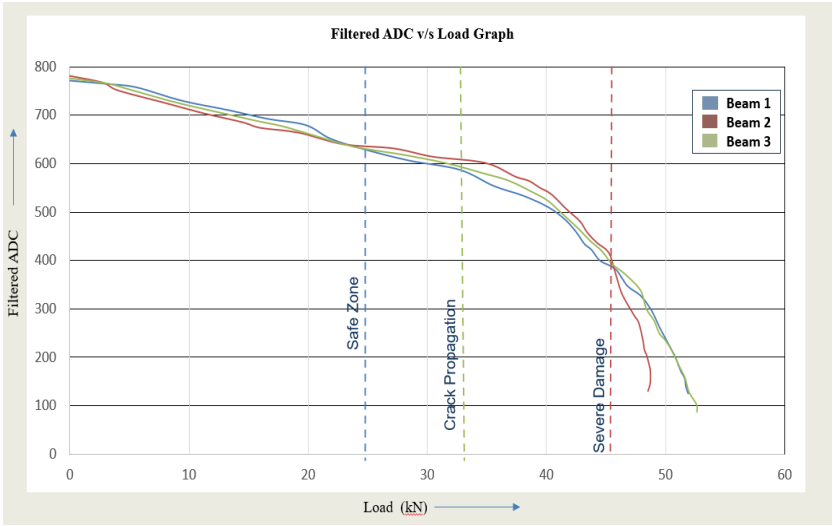

Fig.7 Graphical representation showing ADC vs Load relationship in SFRC

3.2 Load–Deflection and Crack Pattern Analysis

The load-deflection behaviour of the beams is illustrated in Fig.8. RCC beams exhibited a linear response up to 20 kN, followed by a sudden increase in deflection leading to failure within the range of 48-52 kN. This indicates limited deformation capacity and brittle behaviour.

SFRC beams, however, showed a smoother and more gradual transition beyond the cracking stage (after 25 kN), with higher deflection values sustained during the propagation stage (33–43 kN). This reflects improved toughness and energy absorption capacity.

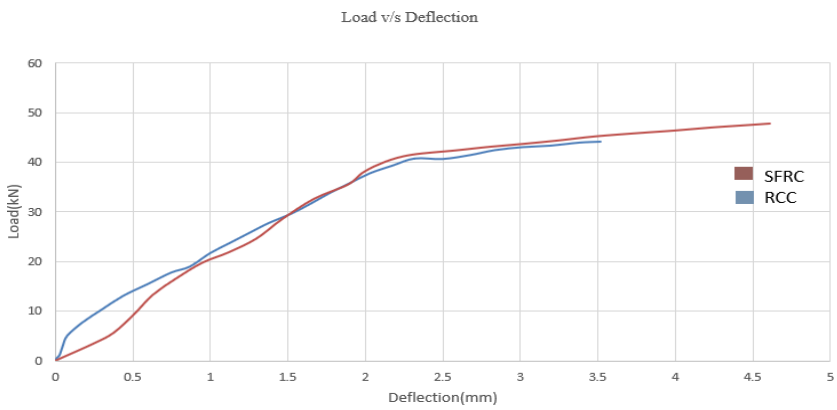

Fig.8 Graphical representation showing Load vs Deflection relationship

The observed crack patterns are presented in Fig.9. RCC beams developed a single dominant crack at mid-span, which widened rapidly with increasing load. In contrast, SFRC beams exhibited multiple fine cracks distributed along the span, indicating effective stress redistribution and crack control due to the presence of steel fibers.

Fig.9 Representation of crack pattern in RCC and SFRC

3.3 Water Ingress Behaviour

The variation of average capacitance with respect to incremental water addition for both Normal Self-Compacting Concrete and Steel Fiber Reinforced Concrete is presented in figure.10. In both cases, capacitance increased with water content, which is attributed to the replacement of air within the pore structure by water, having a significantly higher dielectric constant.

For Normal SCC, the capacitance increased gradually from approximately 100 pF in the dry state to about 200 pF at higher moisture levels. The response remained relatively smooth and linear, indicating uniform pore filling and steady capillary action. At higher water contents, the curve exhibited a plateau, suggesting that the specimen approached saturation, beyond which additional water produced negligible changes in dielectric properties.

In contrast, SFRC exhibited a higher initial capacitance and a markedly nonlinear response. The capacitance increased significantly with water addition, reaching values above 500 pF at saturation. A steep rise in capacitance was observed in the intermediate range of water addition, indicating enhanced moisture interaction within the matrix. This behaviour can be attributed to the presence of steel fibers, which facilitate the formation of conductive moisture pathways, resulting in a percolation-type effect. At higher water contents, the curve gradually stabilized, indicating saturation of the internal structure.

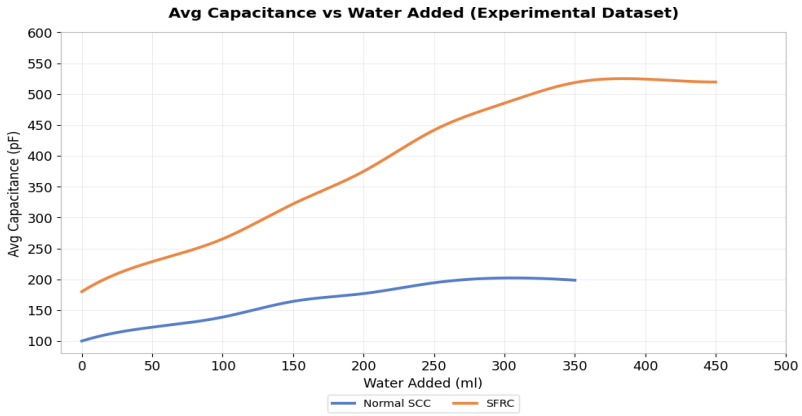

Fig.10 Average Capacitance vs Water Added for SCC and SFRC Specimens.

3.4 Discussion

The crack detection results clearly indicate that SFRC beams exhibit a stable and gradual Load–ADC response without any sudden drop, reflecting controlled crack propagation. The presence of steel fibers acts as crack-bridging elements, preventing abrupt crack opening and promoting the formation of multiple fine cracks instead of a single dominant crack. This leads to a more ductile failure behaviour compared to conventional RCC beams. The sensing system effectively captures this progressive structural response, demonstrating its capability to monitor crack initiation and propagation with high sensitivity.

The water ingress analysis further highlights the enhanced performance of SFRC in comparison to SCC. The significantly higher capacitance variation observed in SFRC indicates a wider sensing range and improved sensitivity to moisture changes. The nonlinear S-shaped response enables clear identification of different stages of water ingress, including surface wetting, internal penetration, and saturation. The plateau region at higher moisture levels serves as a reliable indicator of near-saturation conditions, which is critical for identifying potential durability issues such as reinforcement corrosion.

Overall, the study demonstrates that SFRC outperforms conventional concrete in both structural and durability aspects. The improved ductility, controlled crack propagation, and reduced permeability observed in SFRC, combined with its enhanced sensing response, make it highly suitable for smart structural health monitoring applications. The strong correlation between sensor outputs and physical behaviour confirms the reliability and effectiveness of the proposed monitoring system for real-time damage and moisture detection.

4. Conclusion

This study focused on the development and experimental evaluation of a sensor-based Structural Health Monitoring system for detecting crack formation in Reinforced Cement Concrete and Steel Fiber Reinforced Concrete beams, and water ingress in concrete cubes. The system integrates fiber optic sensing for crack detection and capacitance-based sensing for moisture monitoring, along with real-time data acquisition using an ESP32 and IoT platform.

The experimental results demonstrated the effectiveness of the dual-sensor system in capturing structural response and enabling early damage detection through measurable variations in sensor output.

The main conclusions obtained from the study are as follows:

- Fiber optic sensors effectively detected crack initiation and propagation in RCC and SFRC beams through light intensity variations.
- A clear correlation between applied load and sensor response validated the sensing mechanism.
- RCC beams showed sudden crack propagation and brittle failure, reflected by sharp sensor changes.
- SFRC beams exhibited delayed cracking and gradual failure with smoother sensor response.
- Capacitance sensors successfully monitored water ingress in concrete cubes through changes in dielectric properties.
- Increase in capacitance values with water addition confirmed effective moisture

detection.

- SFRC cubes showed improved resistance to water ingress compared to RCC cubes.
- ESP32 and IoT integration enabled real-time monitoring and data transmission.
- The system provides a non-destructive, cost-effective, and reliable monitoring solution.

Overall, the study confirms that integrating advanced sensing techniques with fiber-reinforced concrete enhances structural monitoring efficiency and durability.

4.1 Future Scope

The study was conducted under controlled laboratory conditions, with scope for further development and field implementation.

- Application to full-scale structural elements such as beams, slabs, and bridges.
- Long-term durability studies under environmental effects such as wet-dry cycles and temperature variations.
- Development of distributed multi-sensor networks for large-scale infrastructure
- Integration with cloud-based analytics for predictive maintenance.
- Incorporation of additional parameters such as strain, temperature, and corrosion.

In conclusion, the proposed system provides a strong foundation for intelligent and sustainable infrastructure monitoring.

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